Sahaj Bhadja

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Summary

Software Engineer and USC MSCS graduate with 2+ years developing reliable production systems. Skilled in **Java, Python**, data structures, algorithms, OOP, REST APIs, microservices, databases, AWS, Linux, testing, and debugging.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Data Structures, Analysis of Algorithms, Operating Systems, Software Engineering

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Database Systems, Computer Networks, Web Technologies

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Restructured fault-tolerant Python workflows with explicit job states, background workers, and failure isolation, tripling throughput for 1K+ concurrent jobs while sustaining 99%+ **operational reliability** across 5K+ monthly tasks.
- Investigated application and database bottlenecks for 500+ users, reducing latency from 1.2 seconds to 350 milliseconds through **performance optimization**, indexed SQL queries, validation checks, code reviews, and debugging.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Developed concurrent Python services for secure file ingestion and multimodal processing, handling 1K+ files per batch with fault isolation, provider fallbacks, and **automated testing** across production workflows.
- Built React and Flask features supporting 150+ instructors and 10K+ daily requests, integrating REST APIs and AWS RDS while **CI/CD deployment checks** reliably reduced feedback turnaround 60%.

USC AD Labs **January 2025 – May 2025**
ML Research Assistant *Los Angeles, CA*

- Built reproducible Python evaluation pipelines processing 100K+ autonomous-driving logs, improving trajectory planning 18% while applying **algorithms and data structures** to controlled simulations, baseline comparisons, and documented results.
- Architected a spatial-temporal decision system combining map geometry, lane context, and ego trajectories, improving closed-loop stability 15% through **system testing**, failure analysis, debugging, and collaborative research reviews.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer – SecureSync *India*

- Engineered cross-platform microservices using **object-oriented design patterns**, improving API response times 30%, reducing technical debt 20%, and strengthening application maintainability across distributed Windows, Linux, and macOS deployments.
- Automated 20+ unit and integration tests with PyTest, Mockito, Jenkins, and virtual-machine builds across three operating systems, supporting **code reviews**, defect resolution, Agile delivery, and reproducible releases.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern – VibrantChem *India*

- Built eight **Java and Spring Boot microservices** for authentication, resources, billing, permissions, and notifications, reliably synchronizing 2M+ records across distributed locations through REST APIs and real-time updates.
- Implemented automated unit, integration, and end-to-end tests through Jenkins CI/CD and Docker staging, achieving 95%+ **build validation** coverage and reducing deployment time 50% across controlled production releases.

Skills

Languages: Java, Python, SQL, C++, TypeScript, JavaScript, C#, Bash, HTML5, CSS3
Backend & Data: Spring Boot, Flask, FastAPI, REST APIs, Microservices, OOP, Design Patterns, PostgreSQL, SQL Server, MySQL, MongoDB, Redis, SQLite, Data Modeling, Query Optimization
Cloud & Systems: AWS, EC2, S3, RDS, Azure, Docker, Kubernetes, Linux, Unix, Concurrency, Multithreading, Distributed Systems, Performance Optimization, System Design
Engineering: Data Structures, Algorithms, Git, GitHub, Code Review, Unit Testing, Integration Testing, PyTest, Mockito, JUnit, CI/CD, GitHub Actions, Jenkins, Agile/Scrum, SDLC, Debugging, Root-Cause Analysis, Technical Documentation, Operational Excellence

Projects

Chat2SQL | *Java, TypeScript, Node.js, Docker, Maven, PostgreSQL, SQL Server*

- Extended a cross-platform **Java developer workspace** supporting 10+ databases, combining schema exploration, query generation, Docker tooling, Maven builds, and reproducible local environments across Windows, macOS, and Linux.

Linguistix | *Python, AWS, Docker, CI/CD*

- Developed and deployed a containerized RESTful data-processing API on AWS using CodePipeline and ECS Fargate, with **automated CI/CD releases**, validation checks, 87% processing accuracy, and production-grade reliability.

CPU-Scheduling Simulator Live | *C++, React.js, JavaScript, EJ-2*

- Implemented FCFS, SJF, SRT, Round Robin, priority, and multilevel-queue scheduling algorithms in C++ with **interactive visualization**, validating 1K+ process combinations and reducing average turnaround time up to 25%.